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CVE 312 — Past Questions & Materials
by Renance

Course: CVE 312
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — CVE 312 STRENGTH OF MATERIALS II 2024I2025

Level 300 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF CIVIL ENGINEERING
FIRST SEMESTER EXAMINATION, 2024/2025 SESSION
CVE 312: STRENGTH OF MATERIALS II UNITS: 2**

Instruction: Answer four (4) Questions only. Time: 2½ Hrs**Question 1**

- (a) A solid shaft is to transmit a torque of 10kN-m. If the shearing stress is not to exceed 45 MPa, find the minimum diameter of the shaft. (12 marks)
- (b) A hollow shaft of external and internal diameters of 80mm and 50mm is required to transmit torque from one end to the other. What is the safe torque it can transmit, if the allowable shear stress is 45Mpa? (13 marks).

Question 2:

A timber beam 15cm wide and 20cm deep carries uniformly distributed load over a span of 4m and is simply supported. If the permissible stresses are 30N/mm^2 longitudinally and 3N/mm^2 transverse shear, calculate the maximum load which can be carried by the timber beam. (25 marks).

Question 3

A uniform cantilever beam 4m long is loaded with a 20kN point load at the free end B and fixed at the left end A. Determine the magnitude and direction of the deflection and slope at point B using the unit load and unit couple methods. Assume that E and I are constant from A to B. (25 marks).

Question 4

A long thin cylinder of thickness 1.0cm and internal diameter 60cm is closed with hemispherical ends. If the poisson ratio (μ) for the material is 0.25 and the maximum stress, $\sigma_{\max}$ within the material is not to exceed 10^5N/m^2 . What is the maximum internal pressure to which the system can be subjected. (25 marks)

Question 5

A rectangular beam 300mm deep is simply supported over a span of 4m. What uniformly distributed load per metre will the beam be able to carry if the bending stress is not to exceed 120N/mm^2 ? Take $I = 8 \times 10^6\text{mm}^4$. (25 marks).

Question 6

A timber beam 100mm wide and 150mm deep supports a uniformly distributed load of intensity $w\text{kN/m}$ -length over a span of 2m. If the safe stresses are 28N/mm^2 in bending and 2N/mm^2 in shear, calculate the safe intensity of the load which can be supported by the beam. (25 marks)